A1 Reproduction and Verification

Commands, environment and what the automatic checks do and do not cover

Step	What	Command (PowerShell, repository root)
1 Environment	Python 3.12.3, CPU; exact run versions in requirements-lock.txt and run.json	`pip install -r requirements-lock.txt -r requirements-timestral.txt`
2 Data (not in Git)	NQ 1-minute bars 2006–2025 (TradeStation, licensed); Timestral-5M weights in the local Hugging Face cache	`$env:FUTURES_DATA_DIR = "C:\path\to\futures-data"`

3 Train and forecast	all models, five timeframes; then the daily timeframe again without the flagged-bar filter, merged with the intraday forecasts (<run> = merged folder)	`python multitarget.py --tag final`
		`python multitarget.py --freqs session --daily-ignore-flags --tag daily_noflags`
		`python tools\merge_daily_run.py <final run> <daily run>`

4 Tables and charts	results, rankings, timeframe map, charts, deck_data.json	`python multitarget_report.py <run> reports\multitarget_final`

5 Sample and Timestral checks	appendix A3, A4, A6 (1-minute data, saved forecasts, Yahoo daily data)	`python tools\sample_checks.py <run> reports\multitarget_final`
		`python tools\timestral_same_sample.py <run> reports\multitarget_final`
		`python tools\yahoo_compare.py data_raw\yahoo reports\multitarget_final presentation`

6 Copy, verify, build	copy deck_data.json, charts and the three JSON files to presentation\	`python verify_multitarget.py <run> reports\multitarget_final presentation` `cd presentation; npm ci; node build_deck.js`

Verified

- every point estimate in the results tables, recomputed independently from the saved forecasts
- the ranking rules, the five-model tables, the timeframe map, the data profile and the deck data
- past-to-future Timestral calibration: no calibration target after the forecasts it is used for
- sample-check content (A3): flags vs threshold and their predecessor, categories, daily accounting vs saved forecasts
- unit tests (pytest) for targets, timing, splits, bootstrap, calibration and flag provenance

Not verified

- the raw 1-minute data and the candle building (measured separately in A3, not re-read by the verifier)
- the training: models were not re-fitted
- bootstrap intervals: only their order is checked
- two runs: intraday forecasts from the 2026-11-12 run, daily forecasts from the 2026-11-14 daily run (no flagged-bar filter)
- the run used an uncommitted version of multitarget.py (hash in run.json); later changes: majority-class fix applied to saved outputs, docstrings

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A2 Selection and Statistics

How models were tuned, ranked and given uncertainty intervals

- **Tuning:** hyper-parameters (Ridge α, logistic C) and early stopping (XGBoost, networks) use the validation block (2017-08-25 → 2021-07-14) only.
- **Evidence score z:** paired validation improvement over the task's naive benchmark (squared-error reduction vs zero; accuracy gain vs majority; squared log-RV error reduction vs persistence) divided by its moving-block bootstrap standard error (20-session blocks, 2,000 replicates).
- **Revision (not blind):** the first volatility score was the QLIKE reduction (Patton, 2011). On validation a few limit-locked March 2020 candles with almost zero variance dominated it, so after a first report had printed test numbers it was replaced by the log-RV squared-error reduction.
- **Five-model tables (slides 6–8):** one model per family on each task's best validation timeframe, ranked by z. Timeframe map (slide 4): the highest-z trained model per task and timeframe.
- **Intervals:** differences (accuracy gains) use the bootstrap of paired per-row differences; ratios (R^2-type error cuts) resample model and benchmark losses together and recompute the ratio in each replicate (Künsch, 1989).
- **Majority-class fix:** the first run counted dojis as "not up"; the majority class was recomputed from non-doji training candles without refitting any model (tools/mt_fix_majority.py).

A3 Sample Checks

Daily sample after removing the flagged-bar filter, and whether incomplete candles matter

A. Daily sample (measured, tools/sample_checks.py)

- **Earlier rule:** a session was excluded if any of the last 50 sessions had a flagged 1-minute bar (|log return vs the previous bar| > 2%): 304 of 1,002 test sessions (50 only because of the 2023-03-16 flag described below), 642 kept, daily test ending 2025-01-30.
- **Now (new daily run):** flagged bars no longer exclude daily sessions (all splits; prices unchanged, intraday run unchanged). 937 of 1,002 test sessions have a forecast, to 2025-05-29. Not included: 64 holiday-shortened sessions or sessions whose next session is shortened, and the last session (no next session).
- **What generated each flag:** computed before out-of-schedule bars are dropped; all 21 flags (16 in schedule) exceed the threshold. 5 in-schedule flags are measured against a dropped 16:59 CT bar 989–991 points off (against the last kept bar < 0.05%); 9 move more than 2% and stay beyond 2% 30 minutes later; 2 partly retrace (2020-03-02, 2020-03-16).
- **Clues, not proof:** persistent moves often coincide with 8:30 ET data releases or a weekend reopening; the 16:59 CT bars look like artefacts. Neither timing nor persistence proves that a price is genuine or wrong.
- **Consequence:** the daily candles keep the flagged minutes, so their realized variance includes those moves. The reconciled daily closing returns of the sessions the earlier rule excluded agree with Yahoo (A6); single minutes and RV are not verified.

B. Full-length candles with missing 1-minute bars

Candle	All years	Test	Test, 8:30–15:00 CT	Test labels affected	DNN 128 error cut (%) all → complete
5 min	18.0%	0.23%	0.003%	601	39.99 → 40.00
15 min	25.8%	0.43%	0.008%	385	35.06 → 35.07
30 min	30.6%	0.65%	0.008%	279	41.72 → 41.80
1 hour	34.3%	1.01%	0.000%	201	41.97 → 42.24

- A full-length candle with 4 of its 5 minutes is accepted as complete; the count is stored (n_minutes) but not used (unit test documents it).
- Missing minutes are mostly overnight in 2006–2017, consistent with thin trading (no bar is written for a minute without trades), and almost absent in regular hours. In the test, a few sessions with long gaps coincide with quarterly contract-roll days (e.g. 2021-09-09, 2022-09-08), consistent with dataset gaps.
- Scope: the SAVED test forecasts rescored without rows whose label candle is incomplete; the rescored metrics (HAR-OLS, DNN, XGBoost) move by at most 0.4 points. Conclusions unchanged in this test-sample check; the effect on training and on the features of kept rows, and a refit on corrected data, were not evaluated.

A4 Timestral on the Same Sample

Mistral's Timestral-5M vs the trained models on exactly the same random test candles (saved forecasts)

Candle	Sample	Return R^2: Timestral	Return R^2: DNN	Direction pp: Timestral, body	Direction pp: Timestral, close-to-close	Direction pp: DNN, body	Vol. error cut: Timestral	Vol. error cut: DNN	Corr. with next log RV: spread / current RV
5 min	20,000	−2.55%	+0.20%	−0.59	−0.66	+0.51	+12.2%	+39.6%	0.76 / 0.76
15 min	20,000	−2.23%	−0.10%	+0.01	−0.08	+1.09	−37.1%	+34.8%	0.74 / 0.84
30 min	20,000	−1.46%	+0.02%	−1.76	−1.88	+0.14	−56.2%	+41.3%	0.71 / 0.84
1 hour	20,000	−1.59%	+0.00%	−2.28	−2.36	+0.02	−74.3%	+41.8%	0.64 / 0.83
1 day	937	−1.94%	+0.43%	−2.03	−1.60	+0.00	−7.4%	+17.2%	0.77 / 0.81

- **Target mismatch is not the reason:** Timestral forecasts the next close, the study scores the candle body; scored on the close-to-close direction it does no better (columns 5–6).
- **Same rows:** the trained models keep their advantage on Timestral's own sample, so the gap is not a sampling artefact.
- **Volatility is indirect:** Timestral's volatility comes from the spread of its PRICE forecast (q10–q90), mapped to log RV by a linear calibration; the trained models forecast log RV directly from past RV. The raw spread correlates with next log RV slightly less than current RV does, and the calibration fails beyond 5 minutes.
- **Scope:** one small model (5M), frozen weights, 512-close context, one market: not a verdict on foundation models in general.
- **Proposed next experiment (not run):** aligned targets — forecast open and close of t+1 (body direction) and the quantiles of the next candle's 1-minute returns, then derive RV from Timestral's own path; compare 28M–209M on the same sample.

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A5 Data Profile

Descriptive statistics of the three targets, validation and test blocks (data_profile.csv)

Block	Candle	Rows	Sessions	UP / DOWN / doji	Return s.d. (bp)	ACF return	Mean log RV	RV rises	ACF log RV
Validation	5 min	269,167	2017-08-25 → 2021-07-14	48.8% / 47.2% / 4.0%	7.0	−0.006	-16.13	48.9%	+0.76
Validation	15 min	88,336	2017-08-25 → 2021-07-14	50.5% / 47.3% / 2.2%	12.0	+0.003	-14.81	47.9%	+0.85
Validation	30 min	44,513	2017-08-25 → 2021-07-14	51.3% / 47.1% / 1.6%	16.3	−0.001	-14.06	46.2%	+0.85
Validation	1 hour	21,642	2017-08-25 → 2021-07-14	52.5% / 46.3% / 1.2%	22.7	+0.011	-13.28	47.3%	+0.84
Validation	1 day	937	2017-08-25 → 2021-07-13	57.6% / 41.9% / 0.4%	121.2	−0.193	-9.58	45.4%	+0.88
Test	5 min	273,340	2021-07-15 → 2025-05-30	49.9% / 48.3% / 1.8%	7.7	−0.005	-15.75	48.9%	+0.75
Test	15 min	90,189	2021-07-15 → 2025-05-30	50.7% / 48.3% / 1.0%	13.3	+0.003	-14.45	47.3%	+0.84
Test	30 min	44,395	2021-07-15 → 2025-05-30	51.1% / 48.1% / 0.8%	18.5	−0.002	-13.69	46.0%	+0.84
Test	1 hour	21,507	2021-07-15 → 2025-05-30	51.5% / 48.0% / 0.5%	26.2	−0.000	-12.90	47.7%	+0.83
Test	1 day	937	2021-07-15 → 2025-05-29	53.0% / 47.0% / 0.0%	137.2	−0.057	-9.06	48.9%	+0.81

ACF = lag-1 autocorrelation (intraday: pairs within a session; daily: consecutive sessions). RV = sum of squared 1-minute log returns inside the candle.

A6 External Comparison: Yahoo Finance

Daily candles vs ^NDX (cash index) and NQ=F (front-month futures, not adjusted): tools/yahoo_compare.py

A. Sessions the earlier filter excluded: daily returns vs Yahoo

Daily sessions	With a quote	Returns agree	Median gap to ^NDX
Test, in 50-session windows (earlier rule)	292 of 304	100.0%	0.034 pp
Test, outside the windows	658 of 698	99.5%	0.030 pp
All splits, in the windows	551 of 569	99.6%	0.032 pp
All splits, outside	4,214 of 4,442	99.7%	0.032 pp

- **Rule:** "agree" = re-levelled close-to-close return within 0.5 pp of ^NDX (15:00 CT close) or of NQ=F (session close). Closing returns only: the range is not in the rule, so the candle, single minutes and RV are not validated.
- **Flagged sessions:** returns agree for 16 of 16 (2023-03-16: +2.67% vs ^NDX +2.66%; 2020-03-16: −13.43% vs ^NDX −13.00%); their ranges are also close to NQ=F (descriptive).
- **Disagreeing:** 13 sessions in 2006–2025 (test: 2023-09-15, 2023-12-18, 2024-12-23), mostly quarterly-expiry days; none was in the earlier 50-session windows.
- **Conclusion:** the reconciled daily returns of the sessions the earlier rule excluded agree with the Yahoo references within the chosen tolerance: the quarantine also removed plausible sessions, and the new daily run includes them (A3). This does not verify the single minutes or the realized volatility; not a data-quality certification.

B. Additive back-adjustment compresses % returns

Period	S.d. ratio vs ^NDX (as used)	S.d. ratio, re-levelled	Correlation, re-levelled
2006–2012	0.42	0.98	0.996
2013–2020	0.74	1.00	0.999
Train + validation	0.59	0.99	0.998
Test	0.90	0.99	0.998

- TradeStation adds every roll gap to earlier prices: the level is about 2,206 points above the unadjusted price in 2010, 771 in 2024. Point changes are kept, percentages are not.
- Re-levelled = a retrospective estimate (offset: centred 21-day median of Yahoo NQ=F minus futures-data), not the contracts' original prices nor an independent validation. After it, daily returns are close to ^NDX (correlation 0.998, beta 0.99).
- Return and log-RV levels and the training history are scaled (log-RV shift ≈ −1.7 in 2006–2012, −0.3 to 0 in the test); the factor changes over time and between training and test. The effect on relative performance was not measured; a fix needs ratio adjustment and a new run.